

VIVEK RAJ SINGH

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EDUCATION

VIT Bhopal University, Bhopal

2023 – 2027

B.Tech - Computer Science and Engineering - **CGPA - 8.70**

Bhopal, Madhya Pradesh

- **Coursework:** Operating Systems, DBMS, Computer Networks, Probability & Statistics, Computer Architecture.

EXPERIENCE

Quantitative Research Intern

May 2026 – Present

Quintal Mind | Remote

India

- Wrote a live options trading engine (**11** versions) on the DhanHQ WebSocket feed with a volatility edge filter, Gemini-powered news sentiment scoring, half-Kelly position sizing, and hard risk rails (daily loss caps, cooldown timers).
- Optimized Python backtesting infrastructure processing **500K+** minute-level NIFTY option-chain rows over **3** years, testing **6** butterfly spreads against P10/P90 bands; mean-reversion signals on these gaps held a **1.8** Sharpe out of sample.
- Validated with a walk-forward train/test split (bands fit on 2 years, traded on 1 unseen year) and live paper trading: **179** trades at **2.49** average risk-reward and **+7.4%** ROI.

PROJECTS

ROCm-Forge (CUDA-to-ROCm AI Copilot)   | Python, QLoRA/PEFT, FastAPI, Docker

May 2026

- Built a **9**-agent migration engine (AMD Developer Hackathon 2026) translating NVIDIA CUDA codebases to AMD ROCm: deterministic Python AST parsing handles core API mapping, LLM agents cover contextual edge cases, cutting manual migration effort by roughly **65%**.
- Fine-tuned CodeLlama-7B with QLoRA (4-bit NF4, rank 16, $\alpha=32$, FP16) on AMD Instinct MI300X GPUs running ROCm 6.2, over a curated dataset of paired CUDA-to-ROCm migration examples.
- Added a hardware-aware scanner catching warp-to-wavefront (**32**→**64** thread) mismatches, Tensor Core-to-MFMA lowerings, and inline PTX, plus a verification pass and build-error copilot; shipped with FastAPI and Docker.

AEGIS (AI Agent Security Gateway)   | FastAPI, React, Gemini

May 2026

- Built an action-layer security gateway that checks every tool call an AI agent proposes against the user's declared intent *before* it executes, blocking indirect prompt injections, data exfiltration, and unauthorized actions that prompt-level filters miss.
- Designed the intent engine on Gemini Flash/Pro to turn user requests into structured authorization boundaries, enforced via Lobster Trap policies with per-action risk scores and a human-review quarantine queue for risky calls.
- Shipped a FastAPI backend and React dashboard with an explanation engine stating why each action was blocked, plus audit-ready compliance exports; live on Vercel.

Stochastic Storage VaR Engine   | PyTorch, statsmodels, FastAPI

Feb 2026

- Built an end-to-end failure-forecasting pipeline on **107.9M** drive-days of Backblaze telemetry (~**312K** drives): an inverse-MAE weighted LSTM + ARIMA ensemble validated with rolling-origin walk-forward CV against naive and damped-ETS baselines.
- Computed $\text{VaR}_{95}/\text{CVaR}_{95}$ failure bounds from **10,000** Gamma-Poisson Monte Carlo paths, with Isolation Forest and 3σ z-score rules catching anomalous fleet days.
- Deployed a FastAPI + Chart.js dashboard on Render with GitHub Actions CI (pytest + end-to-end smoke test on every push) and a schema-generic CSV upload API extending the engine to any fleet.

TECHNICAL SKILLS

Languages: Python, C++, SQL, Rust

ML/DL: PyTorch, TensorFlow, Scikit-learn, statsmodels, NumPy, Pandas, Time-Series Forecasting

LLM & Agents: QLoRA/PEFT Fine-Tuning, Hugging Face Transformers, Multi-Agent Pipelines, Prompt Engineering

Backend & Infra: FastAPI, Docker, Linux, Git/GitHub, Render, CUDA/ROCm

MLOps & Evaluation: GitHub Actions (CI), Walk-Forward Validation, Monte Carlo, Anomaly Detection